REVIEW PACKET / VERSION 1.0 / 07 NOVEMBER 2026

Trading Research Independent quant review

What is built, what it measures, what the evidence says, and what still needs to be proved.

The objective

Develop a repeatable, executable trading process with positive expectancy after all costs and explicit capital-risk limits. The current app supplies research and evaluation infrastructure; it has not established a profitable live strategy.

Built and inspectable	Current evidence / limits
9 dashboard tabs; 85 chart specifications	Public-data research, historical studies, four proxy backtests, paper framework, data-quality gates and a Windows control panel.
Stage 1 implemented and substantially repaired	Independent vendor reconciliation and historical information-vintage validation are incomplete.
Stage 2 evaluation tools partly implemented	No validated consolidated daily thesis, calibrated live sizing, order integration or actual-fill P&L.
Saved market data through 05 Nov 2026	Expected equity session: 06 Nov. Current saved freshness checks fail; automatic sizing is blocked.
Forward paper record: 0 observations	Offline/replay output exists; no persisted forward experiment or decisions in the saved filesystem.

Most consequential result: three of four corrected next-open scenarios lose money over the saved history. The remaining positive scenario has a mean-trade confidence interval that includes zero. Historical normalized-return significance is not evidence of profitable execution.

Review mandate: challenge the data, timing, null models, execution economics and risk assumptions; identify missing evidence and a defensible promotion/rejection protocol. No new market downloads or application changes were made to prepare this packet.

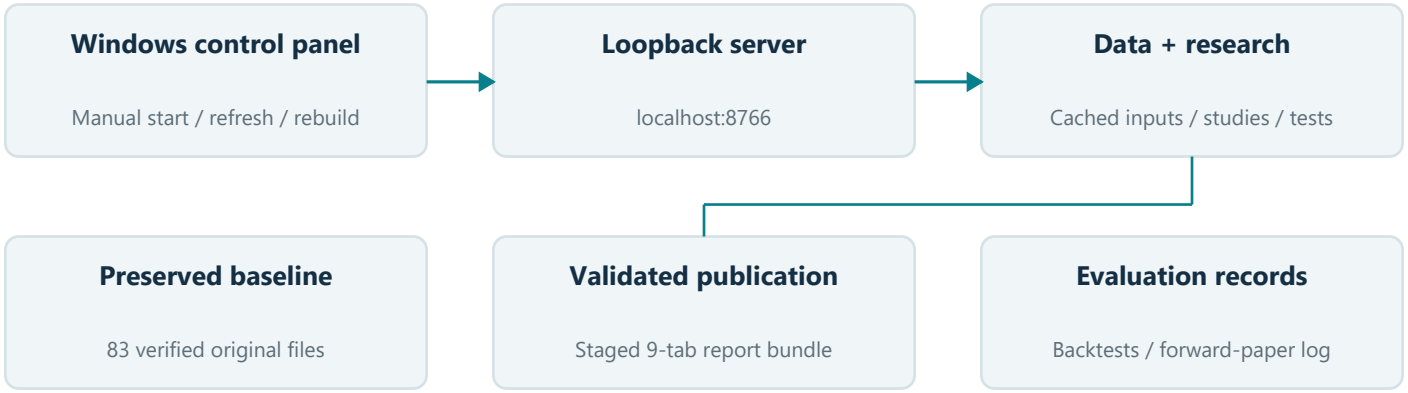
Evidence: [S1-S6]. Saved run: 3dee9a992e9e878f4fd8. Project evidence index: p.17; primary references: p.18.

02 / ORIENTATION

System, scope and goals

How the original research hub became an auditable local application.

The supplied Python app generated four descriptive browser sections: Indexes, Tickers, Leverage and GEX/DIX. The repair work preserved that baseline, corrected calculations and labels, and introduced five evaluation tabs plus native Windows controls. No account orders have been submitted by this app.



Stage	Implemented	Still required
1: describe and verify	Historical datasets, 4 original sections, repaired studies, source/freshness checks, provenance.	Independent price verification; source vintages; dependable current feeds.
2: evaluate a trading process	4 scenario backtests, timing comparisons, uncertainty, forward-log framework, operational controls.	Exact instrument and tradable interval; measured costs; untouched testing; complete forward record; calibrated risk.
Later: controlled deployment	No broker or order execution built.	Only consider after evidence gates, account controls, operational safeguards and separate authorization.

What success should mean

The economic goal is net account growth with acceptable loss risk, rather than a higher chart count, more indicators or a low p-value. A useful model must add value beyond market exposure, survive realistic costs, record abstentions and failures, and remain credible outside the sample used to discover it. Acceptance thresholds are proposed on p.14; they are not yet configured as a complete promotion gate.

Evidence: [S1, S2, S5, S7, S10].

03 / INVENTORY

Data being collected

Saved input coverage and provenance; dates describe this snapshot, not a fresh download.

Input	Coverage / current saved state	Role and boundary
Yahoo adjusted ETF OHLCV	SPY, QQQ, IWM, DIA, RSP, TQQQ, SQQQ, UPRO, SPXU; study history since 2010; ends 05 Nov 2026.	Daily prices/volume and derived research. Adjusted ETF total-return proxies differ from venue perp prices.
Long ETF history	SPY from Jan 1993; QQQ from Mar 1999. Joined long table: 8,500 rows, 20 columns.	Longer risk/tail history; variable start dates and missing coverage matter.
Cboe volatility	VIX, VIX9D, VIX3M, VVIX; saved through 05 Nov. Different series begin in 1990, 2011, 2009, 2006.	Volatility context and conditional studies; no exact next-day return distribution implied.
FRED Treasury yields	DGS2 / DGS10; saved through 04 Nov, expected 05 Nov.	Lagged macro context. Incremental predictive value has not been demonstrated.
Static event calendar	575 entries across 2010-2027: 215 jobs, 215 CPI, 145 Fed; 2 unscheduled Fed decisions.	Event flags and conditional studies. Upcoming dates need source reconciliation.
Kalshi public market APIs	88 saved markets; 86 active. Quotes, risk parameters, funding, hours, volume/OI and hourly candles.	Public observations and short-history tracking; no complete executable historical book.
SqueezeMetrics manual DIX.csv	3,903 observations, 02 May 2011-05 Nov 2026.	Manual positioning import; no automated authenticated download.
BLS and news	BLS release/calendar retrieval code; Yahoo headlines and Fed RSS.	BLS saved retrieval failed; news is descriptive, not a tested sentiment signal.

Data quality is currently blocked

The main daily study table has 4,236 rows and 112 columns (05 Jan 2010-05 Nov 2026). The GEX join has 4,236 rows and 117 columns. All 9 ETFs and 4 Cboe series are one completed session stale in the saved assessment. Arithmetic validation passes, but freshness and independent source truth are different tests.

Not collected comprehensively: true trade/order flow, order-book depth, timestamped historical executable quotes, complete intraday margin paths, first-release macro vintages, pre-release consensus, weekly claims, JOLTS or ADP. These absences limit the economic questions the app can answer.

Evidence: [S2-S4, S8-S9]. Public data access and provider history are dependencies, not guarantees of correctness.

04 / METHODOLOGY

Features and information timing

Precisely what the principal daily studies mean.

Return decomposition and day classification

Close-to-close return uses adjusted ETF closes. Overnight return is prior close to current open; daytime return is current open to current close. Paired decomposition uses the same usable observations, and log-growth shares are labeled explicitly. Missing intermediate sessions no longer compress a multi-session horizon into a one-session label.

Day-type scale

For signal day D, expected daily move = VIX(D-1) / 100 / sqrt(252). Let z = ETF close-to-close return(D) / expected daily move. Categories: big drop z < -1; normal drop -1 <= z < -0.3; quiet -0.3 <= z <= 0.3; normal rise 0.3 < z <= 1; big rise z > 1. This is a volatility-scaling heuristic, not a probability model.

Information	When usable / treatment	Review concern
Completed day type	Requires the signal day closing observation.	An exact same-close fill is not established merely because the historical close exists.
VIX >=20 rule	Uses final signal-day VIX; ordinary scaling uses prior-session VIX.	VIX close availability differs from a 16:00 equity fill. Cboe regular dissemination extends to 16:15 ET. [P1]
GEX / DIX	Historical joins shift the observation one equity session before using it as known.	Conservative lag does not prove the real publication timestamp or its historical vintage.
Treasury yields	Shifted by a session in the daily feature set.	Confirm publication times, revisions and actual use before a macro strategy claim.
Jobs / CPI / Fed flags	Static scheduled dates; emergency Fed decisions separated.	Dates alone contain no actual-versus-consensus surprise information.

Risk and correlation measurements

Historical barrier/tail studies examine daily regular-session high/low excursions from specified reference prices. A threshold touch is neither a stop fill nor an observed liquidation. Correlations and tracking comparisons describe co-movement over overlapping realized intervals; they are not demonstrated forecasts. Daily volume and close-location measures are proxies, not an exchange tape.

Evidence: [S3-S4, S7-S9]; [P1]. No source-vintage proof is implied by lagging a feature or passing prefix checks.

05 / DASHBOARD

The four original research sections

What a reviewer can inspect in the current browser hub.

Tab / chart count	Studies and displayed information	Interpretation
Indexes / 13	Overnight vs daytime growth; VIX calibration; post-day-type follow-through/reversal; index correlations and breakaways; scheduled events; 3x ETF behavior; tails and gaps.	Seven study families. Raw and normalized returns answer different questions; conditional associations need economic testing.
Tickers / 63	9 drill-down pages, 7 charts each: price/history, snapshots, volatility/drawdown, beta/correlation, overnight/day behavior, event effects and adverse excursions. Daily volume/close-location stand-ins plus headlines.	SPY, QQQ, IWM, DIA, RSP, TQQQ, SQQQ, UPRO, SPXU. Descriptive indicators have not each been tested for incremental predictive value.
Leverage / 2	Market inventory, side-specific leverage, proxy barriers, funding sample/coverage, spread, volume/OI and hours. Focused ETF/index tracking and closed-cash-session comparisons.	Short venue history and ETF proxies do not establish a long-run venue strategy or safe leverage. More detail on p.10.
GEX/DIX / 3	Positioning history, trailing DIX rank, GEX/index charts, VIX-conditioned move size and refresh reminders.	Manual input. Hot/Warm/Cold reminders concern freshness, not a directional trade call. Current conditional guidance is suppressed when stale.

Chart integrity versus economic validity

The complete bundle has **85 chart specifications**: 13 Indexes + 63 Tickers + 2 Leverage + 3 GEX/DIX + 4 Backtests. They span 19 HTML files. Chart structure, finite usable values and bundle consistency are checked before publication. This verifies that the report can be rendered consistently; it does not certify every vendor print or the interpretation of a plotted pattern.

What the app does not currently provide

There is no validated single daily up/down/no-edge thesis, portfolio-level capital allocator, real order-flow engine, broker-fill P&L or automatic trade execution. Individual research findings remain visible so that weak and negative results are not hidden.

Evidence: [S1-S4, S8-S10]. Chart totals verified from the saved HTML bundle and desktop verification record.

06 / DASHBOARD

The five added evaluation tabs

Tools introduced to compare, monitor and challenge the original research.

Tab	What is built	Current saved presentation
Original vs fixed	Frozen original reports; old/new calculations run separately on common inputs; repair coverage; full statistical registry.	83 original files verified. Comparisons expose arithmetic/methodology changes, rather than implying a live before/after trading record.
Backtests	4 fixed research candidates; original assumed close entry versus corrected next-open entry; gross/net curves, trade CSV, annual diagnostics, cost sensitivity and uncertainty.	Three corrected scenarios negative; one positive but uncertain. Half/configured/double execution costs; funding held fixed.
Paper monitor	Frozen configuration, before-entry decisions, eligibility/overlap checks, hash-chained events and later outcomes.	Replay mode, 0 events, 0 settled trades. No persisted data/monitor directory or experiment lock in this snapshot.
Data quality	Freshness, arithmetic/structural checks, run provenance, calibration status and source failures.	Validation passes; freshness fails; automatic sizing disabled. No independent vendor certification.
Jobs & macro	Official release/calendar retrieval status; static event reconciliation; imports of timestamp-validated surprises.	Saved BLS requests returned 403; release fields unavailable; zero valid surprises. Upcoming static dates not independently revalidated.

Read the status flags literally

baseline_verified = true confirms preservation. **structural_and_freshness_ok = false** means the combined data gate fails even though arithmetic validation passes. **sizing_allowed = false** is accompanied by: "Broker liquidation model uncalibrated; strategy expectancy not forward validated." A running server is not evidence of usable current data.

The report manifest is dated 07 Nov 2026 and identifies a historical snapshot. Its market-data date remains 05 Nov. A recent build timestamp reflects recalculation of saved inputs, not receipt of a newer market observation.

Evidence: [S2, S5-S6, S8-S10]. Run ID: 3dee9a992e9e878f4fd8.

07 / INFERENCE

Statistical evidence and its limits

The current registry supersedes earlier audit summaries when numbers differ.

The registry lists **33 attempted tests, 31 with finite p-values**. Two month-end tests lack matched controls. Fixed per-test RNG identities use seed 7 and 20,000 Monte Carlo draws. Two-sided departures compare means or log ratios of RMS move size; p = (hits + 1) / (draws + 1), avoiding reported zero p-values.

Individual-day tests resample control days with replacement. Event controls match weekday and exact calendar-gap length, while excluding listed event overlaps and emergency Fed dates from single-event attribution. Multiday tests circular-shift signal dates, preserving clustering/spacing; missing targets can change shifted sample sizes. Exchangeability, stationarity and event-composition assumptions still require review.

Selected test	Raw p	BH q	Holm p
Big-rise next-day direction	.00705	.04371	.19034
Normal-drop five-day direction	.01280	.05173	.31998
Jobs overnight move size	.01150	.05173	.29899
Jobs direction	.69407	.96256	1.00000
CPI overnight move size	.00340	.02635	.09520
Weekend / holiday / negative-GEX size*	.00005	.00052	.00155

*Each of these three size tests has approximately the displayed values; they are separate registry entries. BH and Holm use the 31 finite tests. At 5%, BH retains 5 tests: general big-rise direction, CPI size, weekend size, holiday size and negative-GEX size. Holm retains only the last three size tests. Fed move size does not pass current BH.

Why a p-value is not an account return

Direction tests mostly use VIX-normalized returns, whereas displayed raw means and trading P&L use raw returns. Development-period sign checks (2010-14, 2015-19, 2020-22, 2023 onward) reuse the research sample. "Possible edge" labels use raw p < .05, n >= 30 and agreement in at least 3 eligible periods; they remain exploratory. The VIX >=20 rally split was selected after seeing the overall rally finding.

BH/Holm address the disclosed registry, not every informal search or discarded rule. As a sensitivity example, removing signal dates 15 Feb-30 Jun 2020 reduces the high-VIX rally next-day raw mean from about -0.320% (n=112) to -0.043% (n=98). This is a diagnostic, not an untouched holdout.

Evidence: [S5, S7]. Full registry and source are required to reproduce the null distributions and selection history.

08 / TRADING MODEL

Exactly what the backtests do

Newly introduced hypothetical trading scenarios, rather than a prior bot performance record.

Rule ID / short name	Signal at completed day D	Corrected holding interval
rally_short_1 / Rally 1	Big rise; short	D+1 open to D+1 close
rally_vix20_short_1 / Rally + VIX	Big rise and final VIX >=20; short	D+1 open to D+1 close
normal_drop_long_5 / Drop long 5	Normal drop; long	D+1 open to D+5 close
rally_short_5 / Rally short 5	Big rise; short	D+1 open to D+5 close

The comparison labeled "original" enters at D close and exits at D+h close. It is a simulation introduced during repair using original timing assumptions; the original app had no executable after-cost backtests. Correcting entry to next open changes the economic interval. In particular, a one-session corrected test excludes the overnight move and does not test an executable after-16:30 perp entry.

Configured scenario	Value / treatment
Account and overlap	$10,000 per independent rule; notional fraction 1.0, capped at 1.0. Nonoverlapping positions within each rule; a new signal on the exit date is allowed. No combined portfolio.
Trading friction	1 bp half-spread + 2 bp slippage per side; 2 bp fee per side. Funding 1.36 bp per calendar day; positive funding charges longs and credits shorts.
Accounting	Entry notional = prior equity x fraction. Compounds at exits; daily-close marks reserve hypothetical exit friction/fees. Invalid or incomplete endpoints are skipped.

Return calculation

Let s = +1 long or -1 short; d = (half-spread bps + slippage bps) / 10,000.
Entry fill = entry reference x (1 + s x d); exit fill = exit reference x (1 - s x d).
Price return = s x (exit fill - entry fill) / entry fill.
Fee return = fee bps / 10,000 x (1 + exit fill / entry fill).
Funding return = s x funding bps / 10,000 x elapsed calendar days.
Net return = price return - fee return - funding return.

Funding is a constant prorated scenario, not actual settlement-by-settlement accounting. Fixed 09:30/16:00 ET timestamps omit early closes. There is no order book, fill probability, stop model or liquidation path. Actual venue fees depend on notional, maker/taker status and tier; both legs and funding must be reconciled. [P4-P6]

Evidence: [S6-S7]; [P4-P6]. 1 bp = 0.01%. Four independent account curves cannot be added together.

09 / EVIDENCE

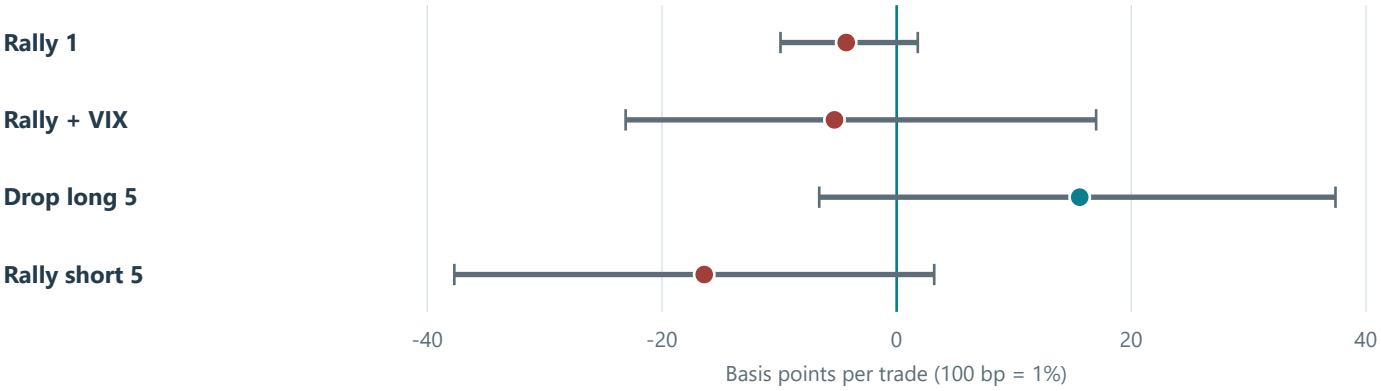
Current historical results

Saved 2010-2026 scenarios; cumulative returns are not annualized.

Rule	Trades	Original gross	Original net	Corrected net	Corrected DD
Rally 1	403	+56.46%	+13.36%	-16.93%	-20.96%
Rally + VIX	112	+40.32%	+28.23%	-6.40%	-10.05%
Drop long 5	451	+390.27%	+100.32%	+82.14%	-21.68%
Rally short 5	299	-36.17%	-36.54%	-42.94%	-50.29%

Each row is a separate, at-most-1x SPY research account. DD is maximum **daily-close marked** drawdown; it is not intraday loss, stop performance or liquidation risk. Original gross/net use same-close entry; corrected net uses next-open entry. Costs remain illustrative ETF-proxy assumptions, not a historical Kalshi execution record.

Mean corrected net return per trade: 95% year-block intervals



Rule	Mean net / trade	95% interval
Rally 1	-0.043%	-0.099% to +0.018%
Rally + VIX	-0.053%	-0.231% to +0.170%
Drop long 5	+0.156%	-0.066% to +0.374%
Rally short 5	-0.164%	-0.377% to +0.032%

All four intervals include zero. The positive Drop long 5 total (+82.14%) is a lead for further testing, not a validated winner. Its long exposure must be compared with an exposure-matched market benchmark, and its selection after viewing these results must be acknowledged.

Intervals use 2,000 seeded resamples of calendar-year trade groups, then the pooled trade mean. They preserve some within-year dependence but are nominal, do not adjust for candidate selection and are not confidence intervals for cumulative account returns. Profit factor uses summed per-entry-notional returns, not dollar profit/loss sums.

Evidence: [S6-S7]. These results are retrospective development diagnostics, not untouched holdout or forward results.

10 / RISK

Venue, leverage, funding and tail risk

Useful public observations; broker-specific risk remains uncalibrated.

The saved Kalshi inventory has 88 markets (86 active): 56 stocks, 20 crypto, 6 metals, 3 indexes, 2 ETFs and 1 energy. Focused active studies cover US500, US Tech 100, SPY and QQQ. US2000 is mapped but inactive in the saved snapshot. US500 and Tech 100 use SPY/QQQ proxies; index-style contracts are not identical ETF instruments.

Product	Published-side leverage estimate L / S	Approx. adverse barrier L / S	Funding history
US500	14.6459x / 12.2355x	1.5757% / 1.8861%	31 total records; trailing 30 over 29 days.
US Tech 100	11.8461x / 10.5644x	1.9481% / 2.1844%	23 records over 22 days.
SPY	4.9628x / 4.9628x	4.6500% / 4.6500%	1 record; extrapolation withheld.
QQQ	4.9628x / 4.9628x	4.6500% / 4.6500%	1 record; extrapolation withheld.

Barrier estimates are not calibrated liquidation prices

At chosen leverage L, the fixed-entry-notional proxy is approximately 1/L - 1/(published side leverage x initial-margin multiplier). The saved multiplier is 1.3. The side-specific table uses each side's own maximum estimate. Mark dynamics, maintenance tiers, size, costs, basis and actual broker implementation are unresolved. The API relationship initial margin = maintenance margin x multiplier does not establish this liquidation formula. [P4-P5]

Saved US500 trailing funding is about $1.2966 per $10,000 notional per day (4.7324% simple annual extrapolation); Tech 100 is about $1.2080 (4.4093%). These short samples end 05 Nov at 16:00 ET. Rates are historical sample summaries, not forecasts or guaranteed carry. Sign, zero rates, timestamp uniqueness and sample completeness are handled explicitly.

Recent tracking examples: US500 has only 22 aligned returns (correlation about .9993); Tech 100 has 16 (about .9971). Overnight endpoint sign agreement is 21/22 and 16/16 on overlapping realized windows. Neither establishes forecastability, entry availability, tail tracking or cost profitability.

Daily ETF high/low barrier touches omit continuous perp prices, market closures, spread widening and intraday account equity. Venue liquidation occurs against maintenance equity and involves market orders. Actual position-level liquidation examples, prices and sizes are needed before using a sizing model. [P5]

Evidence: [S8]; [P4-P6]. Published allowable leverage is not a recommendation to use it. Current sizing remains blocked.

11 / COVERAGE

Jobs reports, macro and positioning

Event dates are covered; numerical surprise signals are not populated.

Jobs reports: what is actually checked

Historical Employment Situation dates are included in the event studies. The app compares prior-close-to-open move size and full-session direction with ordinary days matched on weekday and calendar-gap length. Releases on equity-closed days feed the next available session gap. This measures a broad news window, not an isolated release-timestamp reaction.

Current jobs size n=199 has raw p about .01150, BH .05173; direction raw p about .69407. This is exploratory evidence about move size, not a reliable payroll-direction trading edge. Event count differs from static calendar count because dates outside usable data, overlaps and filtering are excluded.

Capability	Implemented / current state
Official BLS retrieval	Employment Situation page + official calendar fetch/archive code and payroll, unemployment, wage extraction exist. Saved requests both returned HTTP 403 at 06 Nov 07:11:52 UTC.
Numerical releases	No archived release vintage or observed latest release in the app. This is an app retrieval failure; the official Nov 6 release exists. [P2-P3]
Surprise testing	Timestamp-validation import path exists; valid imported surprises = 0. No pre-release consensus feed. A complete initial/revised historical panel has not been assembled.
Other labor indicators	Weekly claims, JOLTS and ADP are not ingested automatically. Participation/revisions and broader release components are not complete tested signals.
Calendar reconciliation	Static jobs/CPI dates extend through 2027, but upcoming dates are not independently revalidated in this snapshot. Fed dates are outside the BLS calendar check.

GEX/DIX and daily order-flow stand-ins

Latest manual positioning: DIX 37.85696%, GEX $1.84475 billion, dated 05 Nov. The app presents rolling ranks and conditional size/risk history; stale current guidance is withheld. Publication times, provider revisions and the one-session availability assumption need verification. Daily relative volume, close location, up-day volume and accumulation/distribution measures are descriptive stand-ins; no actual tape or L2 order flow is collected.

Before testing surprises, retain initial release time/vintage, observation period, revision status, pre-release consensus value and timestamp, and exact tradable reaction window. BLS payroll estimates are revised in subsequent months and annual benchmarking; a latest-value database can introduce look-ahead. [P3]

Evidence: [S3-S5, S9]; [P2-P3]. No unobserved release or consensus values have been fabricated.

12 / CHANGE RECORD

What the audit changed

Repairs improve measurement and expose uncertainty; they do not manufacture an edge.

Area	Repair delivered	Why it matters
Missing margin / side risk	Unknown risk remains unavailable; long and short published inputs stored separately; uncalibrated sizing blocked.	Avoids false zero-hit safety and side-mismatch conclusions.
Return horizons / wealth	Missing sessions retain calendar horizon; valid endpoint wealth; common-sample decomposition; explicit log shares and tiny-dollar precision.	Avoids mislabeled returns, inconsistent compounding and rounded-away losses.
GEX / adverse moves	Proxy-specific adverse highs/lows and side labels; stale current forecasts suppressed.	Avoids applying the wrong instrument or calling stale data current.
Events / volatility	Scheduled vs emergency Fed split; overlap handling; weekday/gap controls; common-sample VIX comparisons.	Reduces known event confounding and sample-composition artifacts.
3x ETFs / tails	Daily-reset compounded benchmark added; absolute tails and threshold-touch labels corrected.	Distinguishes daily leverage from naive multiday multiples; separates touches from fills. [P7]
Funding / tracking	Positive, negative and zero rates separated; coverage/schedule checks; sample/horizon/spread labels.	Avoids extrapolating one record or misreading partial samples.
Statistics	Stable test IDs; fixed 20,000 draws; nonzero Monte Carlo p-values; global raw/BH/Holm registry.	Makes reruns reproducible and disclosed multiple testing visible.
Data and publication	Validate candidate downloads; preserve usable caches; mandatory self-check; high-precision CSV; staged report publication and manifests.	Rejects some corrupted updates and avoids publishing a partial report bundle.
Evaluation / operation	Comparison, backtest, paper, quality and macro tabs; native control panel; preserved baseline and rollback.	Adds reviewable experiment and operation tools, not proof of profitability.

Frozen original HTML remains accessible. Old/new source calculations use common current inputs in separate processes, avoiding accidental source import crossover. Because data and methodology may change, a reviewer should distinguish the frozen original snapshot from a same-input calculation comparison.

Validation boundary: the current report passes arithmetic, day-type, label/coverage and repeated-expiration checks, plus recomputation at six truncated-history cut dates spanning 2013-2026. These prefix checks test transformation consistency when later rows are removed. They do not reconstruct what a vendor actually published at each historical date or eliminate model-selection bias.

Evidence: [S1-S2, S5-S10]. The original Claude workspace was not edited; 83 files were preserved and hash-verified.

13 / REMAINING WORK

Open gaps to close before trading claims

Priority findings from this review, including issues not yet repaired.

Priority / gap	Evidence or limitation	Required next step
P0: source truth and freshness	Saved prices end Nov 5, expected Nov 6; independent vendor reconciliation incomplete. Latest download history is not point-in-time evidence.	Resolve current feeds; reconcile suspect prints/corporate actions; retain raw vintages and discrepancy decisions.
P0: execution interval	Same-close research uses finalized information; next-open repair removes overnight exposure.	Choose one product and actual decision/entry/exit clocks; capture quotes available then.
P0: costs and margin	Constant costs; short funding sample; proxy barriers; daily-close rather than intraday risk.	Measure fees, bid/ask, depth, slippage and funding; calibrate account-level liquidation over price/size examples.
P1: uncertain opens	6 corrected trade rows across 4 dates use SPY opens flagged "open = prior close (unconfirmed)." Descriptive studies filter these; backtests accept finite positive opens.	Verify against independent prices; apply a consistent eligibility policy; rerun sensitivity with exclusions.
P1: early closes	Trading timestamps hardcode 09:30/16:00 ET. Daily close prices may be valid but exit clocks/funding/paper deadlines can be wrong.	Use authoritative session schedules including early closes and product-specific hours.
P1: evaluation design	No untouched chronological holdout; no exposure-matched benchmark, joint capital model or candidate-search adjustment for return intervals.	Freeze protocol; evaluate drift-adjusted value, dependence, multiple selection and portfolio exposure.
P1: forward completeness	0 events; refresh-driven recording; no complete missing-day register, price-vintage freeze or trusted external log anchor.	Begin only after eligibility repairs; record every expected decision, failures and no-trades; archive inputs/configuration.
P2: transaction boundary	Output publication is journaled; raw caches and paper state are outside the same full transaction.	Reconcile durable state after failed builds and preserve provenance linking decisions to the published report.

Flagged entry dates: 2011-12-21, 2016-07-01, 2024-02-08 and 2024-10-16. Flat opens can be genuine; this flag identifies unresolved uncertainty, not a proven bad print. These issues are disclosed in this packet and were not changed during document preparation.

Evidence: [S3, S6-S10]. P0 = economic/data blocker; P1 = validity gap; P2 = operational hardening. Priorities are proposed.

14 / PROFITABILITY PLAN

How we will test for profitability

Proposed evidence protocol for the next stage; acceptance values still need to be frozen.

The test is economic and prospective

Establish positive expectancy after executable costs, under an explicit capital and drawdown budget, with honest uncertainty and a complete forward record. Profit cannot be established by choosing the best historical curve or by watching a few good days.

Gate	Required evidence / measurements	Current state
1. Define and freeze	Instrument, information clock, holding interval, sizing, cost accounting, benchmarks, hypothesis family, minimum meaningful edge and rejection rules.	Partly specified for proxy scenarios. Product/venue execution protocol and promotion thresholds unfinished.
2. Validate historical economics	Clean point-in-time data; feasible entries; measured cost stress; exposure-matched SPY/cash comparisons; dependence-aware diagnostics, then genuinely unseen future or independent tests.	Retrospective scenarios only. All 4 nominal mean-trade intervals include zero; no untouched holdout.
3. Collect forward paper evidence	Every expected before-entry decision, no-trades/missed days, frozen configuration/input snapshots, hypothetical executable fills and actual funding observations.	Framework exists; saved record 0. Manual refresh is a completeness risk.
4. Review a predeclared gate	Positive net expectancy with credible dependence-aware uncertainty; incremental value vs benchmark; acceptable drawdown/tails; cost resilience and stability across periods.	Not yet evaluated. Numeric thresholds, sample size/power and review cadence must be agreed before starting.
5. Reconcile any later live pilot	Only with separate authorization: actual fills/fees/funding vs paper, slippage, outages, capacity and risk-budget adherence.	No order integration or live pilot. Paper outcomes cannot certify live execution.

Scoreboard to build and review

Track net expectancy per trade and per unit time, exposure-adjusted benchmark excess, compounded and annualized return, daily and intraday drawdown, turnover, costs by component, tail loss, concentration, and the fraction of expected decisions recorded on time. Use confidence intervals that account for serial dependence and model selection; measure effective information, not just the trade count.

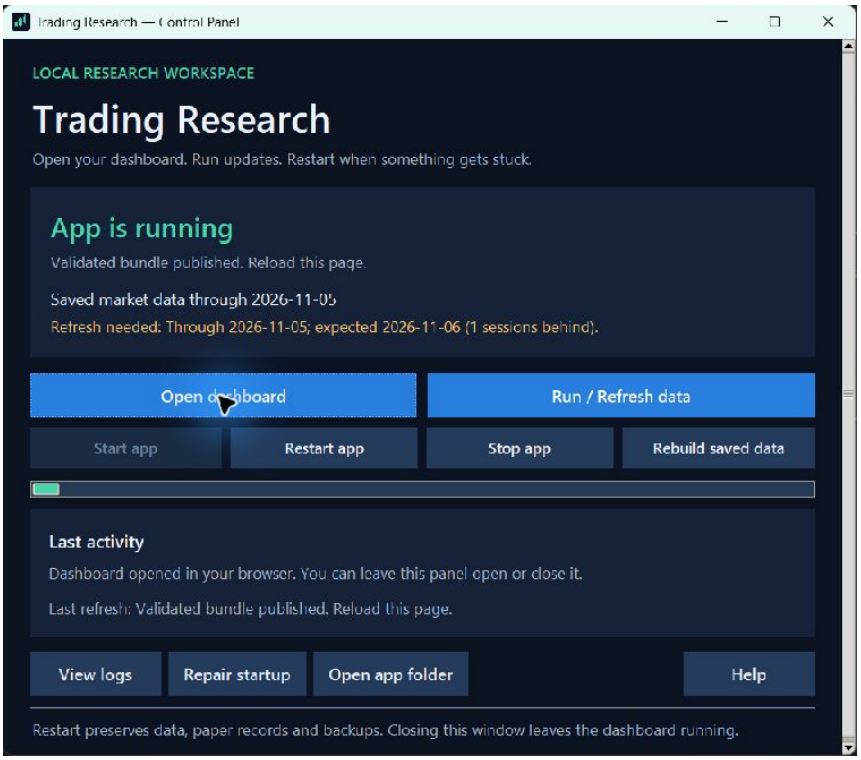
The existing paper framework freezes rules/costs, checks before-entry eligibility, skips overlapping positions and links records by hashes. Offline builds create no forward decisions. Outcomes still use later-downloaded adjusted ETF prices and hypothetical fills. A local hash chain is tamper-evident relative to its chain, not externally immutable. These boundaries must be closed or explicitly retained in the protocol.

Evidence: [S6-S7, S10]; actual fee/funding mechanics require reconciliation to [P4-P6]. This page proposes next work, not completed features.

15 / OPERATIONS

How to run and preserve the app

The Windows control panel makes operation accessible without terminal commands.



Representative native control panel captured during saved desktop verification. Server state and market-data freshness are shown separately.

Action	Behavior
Open dashboard / Start	Start or reuse the identified loopback server, then open the browser. Repeated panel launches reuse one window.
Run / Refresh data	Download current sources and run the validated build. Opening the panel alone does not refresh.
Rebuild saved data	Offline recalculation; does not create new forward paper decisions.
Stop / Restart	Preserves data, reports, paper logs and backups; refuses to interrupt an active or uncertain writer. Closing the panel leaves the server running.
Repair / Logs / Help	Conservative abandoned-lock repair; log viewer, app folder and guide. Refreshes remain manual; no scheduler is installed.

Desktop shortcut: **Trading Research Control Panel**. Browser hub: http://localhost:8766/monitor.html (local to the host running the app). The independent reviewer needs the source/data package to run it on another machine.

Preservation includes original_20261106T064353Z folder/ZIP and before_desktop_controls_v1.zip. Rollback previews by default and makes a recovery archive before apply; default code/report rollback retains current paper logs and market data. Archive before any later experiment or rollback.

Evidence: [S1-S2, S10-S11]. Source ZIP and checksum are listed on p.17.

16 / REVIEW AGENDA

Next work and reviewer questions

Prioritize narrower, falsifiable experiments over adding more indicators.

Sequence	Concrete deliverable
First: data and timing	Fresh, independently reconciled inputs; consistent uncertain-open policy; actual exchange early-close clocks; dependable BLS/calendar handling.
Second: choose one experiment	Name the instrument and entry interval. Investigate Drop long 5 only as a historically selected lead; retain failed candidates and the search record.
Third: model economics and risk	Archive venue quotes, spreads, funding settlements and fee tier; reconcile ETF/perp basis; calibrate maintenance and intraday loss paths.
Fourth: freeze and observe	Predeclare benchmarks, hypotheses, data eligibility, risk/cost thresholds and sample plan; begin complete forward logging with input vintages.
Fifth: review evidence	Compare actual observations with assumptions, measure costs/drawdown/benchmark excess and uncertainty, then accept, extend or reject without hindsight rule edits.

Questions for the independent quant

- What exact instrument, decision time, entry/exit interval and benchmark should define the first test?
- Are adjusted OHLC, corporate actions, suspicious opens and historical calendar vintages trustworthy enough for this economic question?
- Do VIX-normalized hypothesis tests correspond to raw, position-weighted net P&L? What simpler baseline should we beat?
- Do matched controls, circular shifts and year blocks adequately handle regimes, overlap and the full history of searches?
- Can measured execution costs, basis and funding erase the apparent edge? What liquidity/capacity assumptions fail first?
- What intraday gap, tail, stop and liquidation scenarios are missing from daily-close drawdown?
- How should we count missing decisions, no-trades and late records so forward results cannot benefit from selective availability?
- What minimum economic edge, confidence, drawdown budget and effective sample would justify promotion or rejection?
- Which proposed feature adds independently testable value rather than another correlated indicator?
- What evidence would change your mind about the positive candidate, and how should that rejection criterion be recorded now?

Evidence: [S1, S5-S10]. Proposed sequencing is a research plan, not an endorsement of any current strategy.

17 / EVIDENCE INDEX

Project evidence and reproduction

Paths are relative to the trading_research app unless prefixed with package parent.

Ref	Artifact / source	What to verify
S1	Package parent: PROJECT_RECAP_2026-11-07.md; app READ_ME_FIRST.md; CHANGELOG_AUDIT_REPAIR.md	Scope, chronology, original preservation, stage definitions and declared boundaries.
S2	output/build_manifest.json; output/monitor_status.json; output/source_freshness.json; output/validation_report.json	Run identity, hashes, dates, quality gates and transformation checks.
S3	output/daily_table.csv; output/long_history.csv; output/gex_daily.csv; output/data_dictionary.csv; data/raw/*; data/manual/DIX.csv	Observations, definitions, corporate actions, unconfirmed opens, missingness and staleness.
S4	stage1/dataset.py; stage1/studies.py; stage1/sources.py; validation.py	Feature construction, calendar alignment, timing, null models and validation definitions.
S5	output/statistical_registry.json; output/monitor.html; frozen original reports under backups	33 attempted / 31 finite tests; raw/BH/Holm; matched-comparison methodology.
S6	output/backtest_summary.json; output/backtest_trades.csv; output/backtests.html; monitor_config.json	4 separate-account scenarios, fills/costs, sample counts, equity/drawdown and year intervals.
S7	monitor/backtest.py; monitor/paper.py; monitor/build.py; output/paper_state.json	Exact fill math, rule selection, overlap, state persistence, before-entry decisions and zero forward events.
S8	leverage/build.py; output/leverage_markets.csv; data/raw/kalshi/*; data/kalshi/snapshots.csv; output/leverage.html	Published risk inputs, side barriers, funding coverage, tracking and hours.
S9	monitor/employment.py; output/employment_status.json; output/jobs.html; data/events/events.csv; gex/build.py; output/gex_status.json	BLS failures, import validation, event dates, positioning lag and stale-guidance suppression.
S10	build_pipeline.py; app_control.py; desktop_ui.py; restore_baseline.py; CONTROL_PANEL_GUIDE.md; tests/*	Publication boundary, safe server identity/control, process locking, rollback, controls and tests.
S11	Package parent: desktop_verification_2026-11-07.json; trading_research_control_panel.jpg	Saved 59-test pass; start/stop/restart/reuse, offline build, logs, busy-stop refusal and chart validation.

Portable source/data archive: trading_research_desktop_2026-11-07.zip (package parent).
SHA-256: 358372de13f71d0086a43954850e441b27ac7a1dd2fea73d6272ebb477809d16

Reproduce in an isolated copy: verify archive checksum and baseline manifest; follow READ_ME_FIRST; rebuild saved data in offline mode; compare generated manifests/summary values; run the documented test suite. Archive inputs, source and configuration before any online refresh. The 59 saved test passes establish tested behavior, not alpha or provider truth.

18 / REFERENCES

Primary references and review inputs

Official materials checked 07 Nov 2026; application coverage is established by project evidence, not these links.

P1 Cboe - VIX Methodology (v6)
VIX describes option-implied forward volatility, not a guaranteed daily normal-return distribution; dissemination timing matters.
https://cdn.cboe.com/resources/vix/VIX_Methodology.pdf

P2 BLS - Employment Situation schedule and current release
Official scheduled publication source. The Nov 6 release is separately available at the current release URL below.
https://www.bls.gov/schedule/news_release/empsit.htm

P2a BLS - Employment Situation release
A current-release URL changes over time. Archive dated versions; an app HTTP 403 does not establish report unavailability.
https://www.bls.gov/news.release/empsit.nr0.htm

P3 BLS - CES frequently asked questions
Payroll estimates are revised after initial publication and through annual benchmark/seasonal processes; latest values are not first-release vintages.
https://www.bls.gov/web/empsit/cesfaq.htm

P4 Kalshi API - Get risk parameters
Documents risk fields and initial/maintenance multiplier relationship; does not validate the app's full liquidation proxy.
https://docs.kalshi.com/margin-rest/risk/get-risk-parameters

P5 Kalshi - Understanding liquidation
Maintenance equity and liquidation execution matter; static ETF threshold touches are incomplete account-risk evidence.
https://help.kalshi.com/en/articles/15357646-understanding-liquidation

P6 Kalshi - Perps fees explained
Fees use position notional, depend on maker/taker and tier, apply on both legs; funding is separate.
https://help.kalshi.com/en/articles/16071417-perps-fees-explained

P7 ProShares - TQQQ product documentation
A daily 3x target does not imply a 3x multiday cumulative return.
https://www.proshares.com/our-etfs/leveraged-and-inverse/tqqq

Useful information to return with the review

Please identify the target instrument/account type, expected trading interval, fee tier, tolerable drawdown and capital-at-risk budget. If available, provide anonymized timestamped quotes/fills, funding settlements, margin/liquidation examples, independent OHLC and pre-release macro consensus vintages. No account credentials are needed for an initial methodology review.

Requested reviewer output

Return prioritized defects and missing evidence, proposed experiments with falsification criteria, and a concrete acceptance/rejection protocol. Distinguish data correctness, statistical credibility, executable economics and operational readiness so each proposed improvement can be tested.

Prepared from the supplied project and its saved source/data/report bundle. This packet is an audit and research handoff; neither the current application nor this document establishes future profit.